illuminated by a naturalistic clarity in which every residual system of mythic ideas is cleared away, with a great attention to humanity and a deep ethical concern for life – anticipating by some two thousand years the best aspects of the eighteenth-century Enlightenment. The ethical ideal of Democritus is that of a serenity of mind reached through moderation and balance, by trusting in reason and not allowing oneself to be overwhelmed by passions.

Plato and Aristotle were familiar with Democritus's ideas, and fought against them. They did so on behalf of other ideas, some of which were later, for centuries, to create obstacles to the growth of knowledge. Both insisted on rejecting Democritus's naturalistic explanations, in favour of trying to understand the world in finalistic terms – believing, that is, that everything that happens has a purpose; a way of thinking that would reveal itself to be very misleading for understanding the ways of nature – or in terms of good and evil, confusing human issues with matters which do not relate to us.

Aristotle speaks extensively about the ideas of Democritus, and with respect. Plato never cites Democritus, but scholars suspect today that this was out of deliberate choice and not for lack of knowledge of his works. Criticism of Democritus's ideas is implicit in several of Plato's texts, as in his critique of 'physicists', for example. In a passage in his *Phaedo*, Plato has Socrates articulate a reproach to all 'physicists' which will have a lasting resonance. He complains that when 'physicists' had explained that the Earth was round, he rebelled because he wanted to know what 'good' it was for the Earth to be round; how its roundness would benefit it. Plato's Socrates recounts how he had at first been enthusiastic about physics, but had come to be disillusioned by it:

> I had expected to be first told that the Earth was flat or round, but also that, afterwards, the reason for the necessity of this shape would be explained to me, starting from the principle of the best, proving to me that the best thing for the Earth is to have this shape. And if he had said that the Earth was at the centre of the world, then to show me how being at the centre was of benefit to the Earth.[9]

How completely off track the great Plato was here!

Is there a limit to divisibility?

The greatest physicist of the second half of the twentieth century, Richard Feynman, wrote at the beginning of his wonderful introductory lessons on physics:

> If, in some cataclysm, all scientific knowledge were to be destroyed, and only one sentence passed on to the next generation of creatures, what statement would contain the most information in the fewest words? I believe it is the atomic hypothesis, or the atomic fact, or whatever you wish to call it, that *all things are made of atoms – little particles that move around in perpetual motion, attracting each other when they are a little distance apart, but repelling upon being*